英文期末重點整理

U3-A 課文

原文

It’s two in the morning, and a koala is caught on a fence, like a prisoner trying to escape. A phone rings in the home of Megan Aitken in a suburb of Brisbane, on the east coast of Australia. Aitken runs a volunteer organization devoted to rescuing wild koalas. Before she is told the location, she has already thrown her clothes over her pajamas, ready to head out.

When Aitken arrives on the scene, two other volunteers-Jane Davies and Sandra Peachey-are already there. They discover that the koala’s fur is caught in the barbed wire.1 Nearby, they notice tall eucalyptus trees. “He was obviously trying to get to the trees on the other side,” Aitken says.

Aitken puts on heavy gloves. Despite their cute appearance, koalas can be ferocious when resisting capture. If they feel threatened, they bite, and Aitken has the scars to prove it. The volunteers get to work. Davies throws a blanket over the animal, while Peachey opens the lid of a cage. Aitken firmly grasps the koala through the blanket, frees it from the fence, and drops it in the cage.

Next, they check the animal’s physical condition. If the koala is sick or injured, it may need to be taken to an animal hospital. If the koala is healthy—like this one—it is normally released where it is found. Koalas tend to live in a small area, and often eat from the same trees over and over.

Right now, however, Aitken and the rescued koala are in a suburb with almost no trees. “This is the whole problem,” Aitken says. “There are so few places left for the koala.” In the end, Aitken takes the animal to a small park nearby and releases him. “Good luck, little one,” she says.

Koalas at Risk

“Koalas are getting caught in fences and dying,” explains Deidré de Villiers, a koala researcher in Queensland, Australia. Others are being killed by dogs or struck by vehicles, she says. Some even die “simply because a homeowner cut down several eucalyptus trees in his backyard.”

For 15 years, de Villiers has been studying koalas and the reasons for their disappearance. She is also working on ways to make suburban areas more koala-friendly. De Villiers believes that koalas and humans can live together, if certain changes are made. She recommends reducing speed limits on streets and creating more green areas for koalas to live in. Even more important is the need to preserve eucalyptus trees.

Even if these changes are made, koalas still have another problem. “Disease is a huge issue,” explains veterinarian[2] Jon Hanger. Hanger says that almost half of Queensland’s female koalas are affected by a disease called *chlamydiosis*. Without treatment, the koalas are unable to reproduce. “Koala populations that used to be vibrant are becoming extinct,” says Hanger. Once, there were millions of koalas in Australia; now, there are believed to be fewer than 80,000.

A Friend to Koalas

At her home near Brisbane, Deidré de Villiers is taking care of a female koala named Ruby. "Ruby still sleeps in the basket hugging her teddy bear," she says. "She was rescued from the jaws of a dog." Every two days, de Villiers collects eucalyptus leaves, the koala's primary food, from a nearby farm to feed Ruby. For 12 years, she has cared for more than 60 koalas.

Later, de Villiers visits a forest near Brisbane to catch Tee Vee, a wild female koala. De Villiers has been following Tee Vee for over a year. Using special audio equipment, de Villiers walks and listens for a signal from the koala's radio collar.[3] She eventually finds Tee Vee sitting on a tree branch 15 meters high. As de Villiers climbs up a ladder, Tee Vee starts moving down the tree. Then, suddenly, the koala jumps into the air and lands on the ground. She is quickly captured by de Villiers's team.

Tee Vee is given medicine to relax her. Next, de Villiers measures the length of the koala's body and head. She also checks Tee Vee's teeth and the condition of her fur. "I think she has a baby," de Villiers says. She carefully opens the koala's pouch and takes out a 10-centimeter-long baby koala. De Villiers examines the baby for any problems. Then she puts it back in the mother's pouch. "As long as there are healthy babies," she says, "there's still hope."

翻譯：

凌晨兩點，一隻無尾熊被困在圍欄上，就像一名試圖逃跑的囚犯。電話在梅根·艾特肯（Megan Aitken）位於澳洲東海岸布里斯本郊區的家中響起 。艾特肯經營著一個致力於營救野生無尾熊的志工組織。在得知地點之前，她就已經在睡衣外套上衣服，準備出發了。

當艾特肯到達現場時，另外兩名志工——簡·戴維斯和桑德拉·皮奇——已經在那裡了。她們發現無尾熊的毛被刺網纏住了。在附近，她們注意到高大的尤加利樹。「他顯然是想去另一邊的樹林，」艾特肯說。

艾特肯戴上厚手套。儘管無尾熊外表可愛，但在抵抗捕獲時可能會非常兇猛 。如果牠們感到受到威脅，就會咬人，而艾特肯身上的傷疤就是證明 。志工們開始工作。戴維斯將毯子蓋在動物身上，而皮奇則打開籠子的蓋子。艾特肯隔著毯子牢牢地抓緊無尾熊，將其從圍欄上解救出來，並放進籠子裡 。

接著，她們檢查動物的身體狀況 。如果無尾熊生病或受傷，可能需要被送往動物醫院。如果無尾熊很健康——就像這隻一樣——通常會被放回發現牠的地方。無尾熊傾向於生活在一個小區域內，並且經常反覆吃同樣幾棵樹的葉子 。

然而，現在艾特肯和獲救的無尾熊正處於一個幾乎沒有樹木的郊區 。「這就是問題所在，」艾特肯說。「留給無尾熊的地方太少了。」最後，艾特肯將動物帶到附近的一個小公園並釋放了牠。「好運，小傢伙，」她說。

處於危險中的無尾熊
「無尾熊被圍欄困住並死亡，」澳洲昆士蘭的無尾熊研究員戴德瑞·德·維利爾斯解釋道。她說，其他無尾熊則是被狗咬死或被車輛撞擊。有些甚至「僅僅因為屋主砍掉了後院的幾棵尤加利樹」而死亡。

15 年來，德·維利爾斯一直在研究無尾熊及其消失的原因。她還致力於尋找讓郊區對無尾熊更友善的方法。德·維利爾斯相信，如果做出某些改變，無尾熊和人類是可以共存的。她建議降低街道限速，並為無尾熊創造更多綠地。更重要的是保護尤加利樹的必要性。

即使做出了這些改變，無尾熊仍面臨另一個問題。「疾病是一個巨大的問題，」獸醫喬恩·漢格解釋道。漢格說，昆士蘭近一半的雌性無尾熊受到一種名為披衣菌病的疾病影響。如果沒有治療，無尾熊將無法繁殖 。

「曾經充滿活力的無尾熊族群正走向絕種，」漢格說 。澳洲曾有數百萬隻無尾熊；現在據信少於 8 萬隻。

無尾熊之友
在布里斯本附近的家中，戴德瑞·德·維利爾斯正在照顧一隻名叫露比的雌性無尾熊。「露比睡覺時仍會抱著籃子裡的泰迪熊，」她說。「她是從狗嘴裡救出來的。」德·維利爾斯每兩天會從附近的農場收集尤加利葉（無尾熊的主要食物）來餵養露比 。12 年來，她已經照顧了 60 多隻無尾熊。

後來，德·維利爾斯前往布里斯本附近的一片森林去捕捉野生雌性無尾熊 Tee Vee。德·維利爾斯已經追蹤 Tee Vee 一年多了。利用特殊的音訊設備，德·維利爾斯一邊走一邊聽取無尾熊無線電項圈發出的信號 。她最終發現 Tee Vee 坐在 15 公尺高的樹枝上。當德·維利爾斯爬上梯子時，Tee Vee 開始往樹下移動。接著，這隻無尾熊突然跳入空中並落在地上。她很快就被德·維利爾斯的團隊捕獲了 。

團隊給了 Tee Vee 藥物讓她放鬆 。接著，德·維利爾斯測量了這隻無尾熊的頭部與身體長度 。她還檢查了 Tee Vee 的牙齒以及皮毛狀況 。「我想她有寶寶了，」德·維利爾斯說 。她小心地打開無尾熊的育兒袋，取出了一隻 10 公分長的無尾熊寶寶 。德·維利爾斯檢查了寶寶是否有任何問題 。然後，她將寶寶放回母親的育兒袋中 。「只要還有健康的寶寶，」她說，「就還有希望。」

U3-B 課文

原文

“When a snow leopard stalks its prey among the mountain walls, it moves . . . softly, slowly,” explains Indian biologist Raghunandan Singh Chundawat, who has studied the animal for years. “If it knocks a stone loose, it will reach out a foot to stop it from falling and making noise.” One might be moving right now, perfectly silent, maybe close by. But where?

Best known for its spotted coat and long distinctive tail, the snow leopard is one of the world’s most secretive animals. These elusive[1] cats can only be found high in the remote, mountainous regions of Central Asia. For this reason, and because they hunt primarily at night, they are very rarely seen.

Snow leopards have been officially protected since 1975, but enforcing this law has proven difficult. Many continue to be killed for their fur and body parts, which are worth a fortune on the black market.[2] In recent years, though, conflict with local herders has also led to a number of snow leopard deaths. This is because the big cats kill the herders’ animals, and drag the bodies away high up in the mountains to eat.

As a result of these pressures, the current snow leopard population is estimated at only 4,000 to 7,000, and some fear that the actual number may already have dropped below 3,500. The only way to reverse this trend and bring these cats back from their threatened status, say conservationists, is to make them more valuable alive than dead.

A Fragile Relationship

Because farming is difficult in Central Asia’s cold, dry landscape, traditional cultures depend mostly on livestock (mainly sheep and goats) to survive in these mountainous regions. At night, when snow leopards hunt, herders’ animals are in danger of snow leopard attacks. If a family loses even a few animals, it can push them into desperate poverty. “The wolf comes and kills, eats, and goes somewhere else,” said one herder, “but snow leopards are always around. They have killed one or two animals many times . . . Everybody wanted to finish this leopard.”

To address this problem, local religious leaders have called for an end to snow leopard killings, saying that these wild animals have the right to exist peacefully. They’ve also tried to convince people that the leopards are quite rare, and thus it is important to protect them.

The Value of Preservation

Financial incentives are also helping to slow snow leopard killings. The organization Snow Leopard Conservancy India Trust has established Himalayan Homestays, a program that sends visitors to herders’ houses. For a clean room and bed, meals with the family, and an introduction to their culture, visitors pay about ten U.S. dollars a night. If guests come once every two weeks through the tourist season, the herders will earn enough income to replace the animals lost to snow leopards. In addition, the organization helps herders build protective fences that keep out snow leopards. It also conducts environmental classes at village schools, and trains the organization's members as nature guides, available for hire. In exchange, the herders agree not to kill snow leopards.

In Mongolia, a project called Snow Leopard Enterprises (SLE) helps herder communities earn extra money in exchange for their promise to protect the endangered cat. Women in Mongolian herder communities make a variety of products—yarn for making clothes, decorative floor rugs, and toys—using the wool from their herds. SLE buys these items from herding families and sells them abroad. Herders must agree to protect the snow leopards and to encourage neighbors to do the same.

The arrangement increases herders' incomes by 10 to 15 percent, and elevates the status of the women. If no one in the community kills the protected animals over the course of a year, the program members are rewarded with a 20 percent bonus in addition to the money they've already made. An independent review in 2006 found no snow leopard killings in areas where SLE operates. Today, the organization continues to add more communities.
Projects like the Homestays program in India and SLE's business in Mongolia are doing well. Though they cover only a small part of the snow leopard's homeland, they make the leopards more valuable to more people each year. If these programs continue to do well, the snow leopard may just have a fighting chance.

翻譯：
雪豹：山中的隱者
「當雪豹在山壁間跟蹤獵物時，牠的動作……輕柔且緩慢，」研究雪豹多年的印度生物學家拉古南丹·辛格·楚達瓦特（Raghunandan Singh Chundawat）解釋道。「如果牠踢落了石頭，牠會伸出一隻腳擋住它，防止它掉落發出聲音。」此時此刻，可能就有一隻雪豹正完美地保持沉默，或許就在附近。但牠在哪裡呢？

雪豹以其斑點皮毛和長而特出的 (distinctive) 尾巴聞名，是世界上最神祕的動物之一 。這些行蹤隱密的貓科動物只能在中亞偏遠的高山地區被發現。由於這個原因，加上牠們主要在夜間狩獵，因此極少被人看見。

生存威脅與衝突
雪豹自 1975 年起就受到正式地 (officially) 保護，但執行 (enforcing) 這項法律已被證明相當困難 。許多雪豹仍因其皮毛和身體部位被殺害，這些東西在黑市上價值連城。然而近年來，與當地牧民的衝突 (conflict) 也導致了許多雪豹死亡 。這是因為這些大貓會殺死牧民的牲畜，並將屍體拖 (drag) 到高山上食用 。

受這些壓力影響，目前雪豹族群數量估計僅剩 4,000 到 7,000 隻，有人擔心實際數量可能已降至 3,500 隻以下。環保人士表示，要逆轉 (reverse) 這種趨勢並讓這些貓科動物脫離受威脅的地位 (status)，唯一的辦法就是讓牠們活著比死掉更有價值 。

脆弱的關係

由於中亞寒冷乾燥的地貌 (landscape) 難以耕種，傳統文化主要依賴牲畜（主要是綿羊和山羊）在這些山區生存 。在雪豹狩獵的夜晚，牧民的牲畜面臨被攻擊的危險。如果一個家庭失去了哪怕幾隻動物，都可能使他們陷入絕望的貧窮 (poverty) 。「狼來了，殺了吃掉後就去別的地方，」一位牧民說，「但雪豹總是在附近。牠們多次殺死一兩隻動物……每個人都想解決掉這隻豹。」

為了處理這個問題，當地的宗教領袖呼籲停止殺害雪豹，表示這些野生動物有權和平生存。他們也試圖說服人們雪豹非常罕見，因此保護牠們很重要。

保育的價值

經濟誘因也有助於減少殺害雪豹的行為。「印度雪豹保護信託組織」建立了「喜馬拉雅民宿」計畫，將遊客送往牧民家中。遊客支付每晚約 10 美元，可獲得乾淨的房間、床位、與家庭共進餐點並體驗其文化。如果在旅遊季節每兩週有一位客人入住，牧民獲得的收入就足以彌補被雪豹吃掉的牲畜損失。此外，該組織還幫助牧民建造防止雪豹進入的保護性圍欄。

在蒙古，一項名為「雪豹企業」(SLE) 的計畫幫助牧民社區賺取額外的 (bonus) 金錢，以換取他們保護這種瀕危貓科動物的承諾 。社區婦女利用自家牲畜的羊毛製作毛線、地毯和玩具，SLE 購買這些產品並外銷。這項安排使牧民收入增加了 10% 到 15%，並提升了女性的地位 (status) 。如果該社區一年內沒有人殺害受保護動物，計畫成員除了已賺到的錢之外，還能獲得 20% 的獎金 (bonus) 。

像印度的民宿計畫和蒙古的 SLE 業務進展順利。雖然這些計畫僅覆蓋雪豹家園的一小部分，但牠們讓雪豹每年對更多人產生價值。如果這些計畫持續成功，雪豹或許就有一線生機。

U3-A		
capture	捕獲	追逐 跟蹤 to catch an animal after chasing or following it
extinct	絕種的	存在 no longer in existence
grasp	緊抓	握 穩 hold firmly
physical	身體的	相關 related to someone’s body rather than their mind or emotions
primary	主要的	地位 of first rank or important or value
reproduce	繁殖	後代 to have offspring
signal	信號	a sound or an action that you make in order to give information to someone
suburb	郊區	遠離 an area where people live which is away from the center of a town or city
tend to	傾向於	which happens often and is likely to happen again
threaten	威脅	構成 pose a threat to
U3-B		
bonus	獎金	額外 an extra amount of money that is given to you as a present or reward
conflict	衝突	分歧 an active disagreement between people
distinctive	有特色的	having a quality or characteristic that makes a person or thing different from others : different in a way that is easy to notice
Drag	拉、拖	to pull something behind you
Enforce	執行	守法 to make people obey a law
landscape	風景	the way a large area of countryside looks (hilly, flat, etc.)
officially	正式地	公開地 正式地 publicly or formally
Poverty	貧窮	非常 the condition of being extremely poor
reverse	往反向移動	反向 to go in the opposite direction

<table>
<tr><td>status</td><td rowspan="2">位 置
high position or rank in society</td></tr>
<tr><td>地位</td></tr>
</table>

U4-A 課文

原文：

Volcanoes are creators and destroyers. They can shape lands and cultures, but can also cause great destruction and loss of life. Two of the best known examples are found at opposite ends of the world, on the Pacific Ring of Fire.

Symbol of Japan

It’s almost sunrise near the summit of Japan’s Mount Fuji. Exhausted climbers, many of whom have hiked the 3,776 meters through the night to reach this point, stop to watch as the sun begins spreading its golden rays across the mountain. For the climbers, this is an important moment. They have witnessed the dawn on Mount Fuji—the highest point in Japan.

Mount Fuji is a sacred site. Japan’s native religion, Shintoism, considers Fuji a holy place. Other people believe the mountain and its waters have the power to make a sick person well. For many, climbing Fuji is also a rite of passage. Some do it as part of a religious journey; for others, it is a test of strength. Whatever their reason, reaching the top in order to stand on Fuji’s summit at sunrise is a must for many Japanese. Every July and August, hundreds of thousands attempt to do so.

Fuji is more than a sacred site and tourist destination, however. It is also an active volcano around which four million people have settled, and it sits just 112 kilometers from the crowded streets of Tokyo. The last time Fuji erupted, in 1707, it sent out a cloud of ash that covered the capital city and darkened the skies for weeks.

Today, new information has some volcanologists concerned that Fuji may soon erupt again. According to Motoo Ukawa and his associates at the National Research Institute for Earth Science and Disaster Prevention, there has been an increase in activity under Fuji recently. This activity may be caused by low-frequency earthquakes. Understanding what causes these quakes may help scientists predict when Fuji will come back to life. In the meantime, locals living near Fuji hold special festivals each year to offer gifts to the goddess of the volcano—as they have for generations—so that she will not erupt and destroy the land and its people below.

Mexico’s Threatening Mountain

Halfway across the globe from Fuji, Popocatépetl—one of the world’s tallest and most dangerous active volcanoes—stands just 70 kilometers southeast of Mexico City. Although the volcano (whose name means “smoking mountain”) has erupted many times over the centuries, scientists believe its last great eruption occurred around AD 820. In recent years, Popocatépetl is once again threatening the lives of the people near the mountain; in December 2000, almost 26,000 people were evacuated when El Popo—as Mexicans call the mountain—started to send out ash and smoke. As with all active volcanoes, the question is not *if* it will erupt again (an eruption is inevitable); the question is *when* it will happen.

“Every volcano works in a different way,” explains Carlos Valdés González, a scientist who monitors El Popo. “What we’re trying to learn here are the symptoms signaling that El Popo will erupt.” These include earthquakes, or any sign that the mountain’s surface is changing or expanding. The hope is that scientists will be able to warn people in the surrounding areas so they have enough time to evacuate.[1] A powerful eruption could displace over 20 million people—people whose lives can be saved if the warning is delivered early enough.

For many people living near El Popo—especially farmers—abandoning their land is unthinkable. As anyone who farms near a volcano knows, the world’s richest soils are volcanic. They produce bananas and coffee in Central America, fine wines in California, and enormous amounts of rice in Indonesia.

Today, many people continue to see El Popo as their ancestors did. According to ancient beliefs, a volcano can be a god, a mountain, and a human all at the same time. To appease[2] El Popo and to ensure rain and a good harvest, locals begin a cycle of ceremonies that starts in March and ends in August. Carrying food and gifts for the volcano, they hike up the mountain. Near the summit, they present their offerings, asking the volcano to protect and provide for one more season.

翻譯：
火山既是創造者也是破壞者。它們能塑造土地與文化，但也能造成巨大的破壞 (destruction) 與生命損失 。其中兩個最著名的例子位於世界兩端，分別座落在太平洋火環帶上。

日本的象徵
在日本富士山山頂 (summit) 附近，現在接近日出時分 。筋疲力盡的登山者們——其中許多人為了抵達這裡，連夜爬升了 3,776 公尺——停下腳步，看著太陽開始將金色的光芒撒向山脈。對於登山者來說，這是一個重要時刻。他們目擊 (witness) 了富士山上的黎明 (dawn)——那是日本的最高點 。

富士山是一個神聖的地點。日本的本土宗教「神道教」將富士山視為聖地。其他人則相信這座山及其水源具有治癒病人的力量。對許多人來說，攀登富士山也是一種成年禮。有些人將其作為宗教旅程的一部分；對其他人來說，這是一場體力的挑戰。無論出於何種原因，在日出時分站在富士山山頂 (summit) 是許多日本人的必做清單 。每年七月和八月，都有數十萬人嘗試挑戰。

然而，富士山不僅僅是聖地和觀光景點。它也是一座活火山，周邊居住著 400 萬居民，距離擁擠的東京街頭僅 112 公里。富士山最後一次噴發是在 1707 年，當時噴出的火山灰雲覆蓋了首都，讓天空連續數週暗無天日。

今天，新的資訊讓一些火山學家感到擔憂，認為富士山可能很快會再次噴發。根據國立防災科學技術研究所（National Research Institute for Earth Science and Disaster Prevention）的宇川元（Motoo Ukawa）及其同事的研究，富士山地下的活動近期有所增加 。這些活動可能是由低頻地震引起的。了解這些地震的原因可能有助於科學家預測富士山何時會「復活」。同時，住在富士山附近的居民每年會舉行特別的祭典，向火山女神獻上禮物——就像他們世世代代所做的那樣——祈求

她不要噴發，摧毀下方的土地與人民。

墨西哥的威脅之山

在地球的另一端，波波卡特佩特火山——世界上最高且最危險的活火山之一——矗立在墨西哥城東南僅 70 公里處。雖然這座火山（名字意為「冒煙的山」）在幾個世紀以來噴發過多次，但科學家認為其最後一次大噴發發生在公元 820 年左右。近年來，波波卡特佩特火山再次威脅到附近居民的生命；2000 年 12 月，當「El Popo」（墨西哥人對這座山的暱稱）開始噴出灰燼與煙霧時，有近 26,000 人撤離。與所有活火山一樣，問題不在於它是否會再次噴發（噴發是不可避免的 (inevitable)），問題在於何時會發生 。

「每座火山的運作方式都不同，」監控 (monitor) El Popo 的科學家卡洛斯·瓦爾德斯·岡薩雷斯（Carlos Valdés González）解釋道 。 「我們在這裡試圖學習的是預示 El Popo 將要噴發的徵兆。」這些徵兆包括地震，或任何山體表面正在改變或擴張 (expand) 的跡象 。科學家希望能夠警告周邊地區的人們，讓他們有足夠的時間撤離。一次強大的噴發可能會迫使 (displace) 超過 2,000 萬人遷離——如果警告發布得夠早，這些人的生命就能獲救 。

對於許多住在 El Popo 附近的居民（特別是農民）來說，放棄他們的土地是不可想像的。任何在火山附近務農的人都知道，世界上最肥沃的土壤是火山灰土壤。它們在美洲中部出產香蕉和咖啡，在加州產出美酒，並在印尼產出大量的稻米。

今天，許多人仍像他們的祖先 (ancestor) 一樣看待 El Popo 。根據古老的信仰，火山可以同時是神、山與人。為了安撫 El Popo 並確保降雨與豐收，當地人會開始一系列的儀式，從三月開始到八月結束。他們帶著要送給火山的食物與禮物爬上山，在接近山頂 (summit) 的地方獻上祭品，祈求火山在接下來的一個季節裡繼續保護並供養他們 。

U4-B 課文

原文：

Never before have so many people been packed into cities—places such as Los Angeles, Istanbul, Tokyo, and Lima—that are regularly affected by earthquakes. Located near the edge of Earth’s huge, shifting plates, these cities face the risk of serious damage and economic disaster from large quakes—as well as the tsunamis, fires, and other kinds of destruction they often cause.

We understand earthquakes better than we did a century ago. Scientists would like to be able to predict them, but is this possible? Today, some of the simplest questions about earthquakes are still difficult to answer: Why do they start? What makes them stop? Perhaps the most important question scientists need to answer is this: Are there clear patterns in earthquakes, or are they basically random and impossible to predict?

In Japan, government scientists say they have an answer to the question. “We believe that earthquake prediction is possible,” says Koshun Yamaoka, a scientist at the Earthquake Research Institute at the University of Tokyo. In fact, Japan has already predicted where its next great earthquake will be: the region of Tokai southwest of Tokyo. Here, two plate boundaries have generated huge earthquakes every 100 to 150 years, but there hasn’t been a major quake here since 1854. The theory is that stress is building up in this zone, which could lead to a massive quake. Unfortunately, this is more a forecast than a prediction. It’s one thing to say that an earthquake is likely to happen in a high-risk area. It’s another to predict exactly where and when the quake will occur.

The desire for a precise prediction of time and place has led to another theory: the idea of “preslip.” Naoyuki Kato, a scientist at the Earthquake Research Institute, says his laboratory experiments show that before a fault in the Earth’s crust finally breaks and causes an earthquake, it slips[1] just a little. If we can detect these early slips taking place deep in the Earth’s crust, we may be able to predict the next big quake.

Clues in the Desert

Scientists working in Parkfield, California, are also trying to see if predicting earthquakes is possible. They’ve chosen the town of Parkfield not only because the San Andreas Fault runs through it, but because it’s known for having earthquakes quite regularly— approximately every 22 years. In the late 1980s, scientists in Parkfield decided to study the fault to see if there were any warning signs prior to a quake. To do this, they drilled deep into the fault and set up equipment to register activity. Then they waited for the quake.

Year after year, nothing happened. When a quake did finally hit on September 28, 2004, it was years off schedule, but most disappointing was the lack of warning signs. Scientists reviewed the data but could find no evidence of anything unusual preceding the quake. It led many to believe that perhaps earthquakes really are random events. Instead of giving up, though, scientists in Parkfield dug deeper into the ground. By late summer 2005, they had reached the fault’s final depth of three kilometers, where they continued collecting data, hoping to find a clue.

And then they found something. In an article published in the July 2008 journal *Nature*, the researchers in Parkfield claimed to have detected small changes in the fault shortly before an earthquake hit. What had they noticed? Just before a quake, the cracks in the fault had widened slightly. Scientists registered the first changes 10 hours before an earthquake of 3.0 on the Richter scale[2] hit; they identified identical signs two hours before a 1.0 quake— demonstrating that perhaps the "preslip" theory is correct. In other words, it may in fact be possible to predict an earthquake.

Although there is still a long way to go, it appears from the research being done all over the world that earthquakes are not entirely random. If this is so, in the future we may be able to track the Earth's movements and design early-warning systems that allow us to predict when a quake will happen and, in doing so, prevent the loss of life.

BUILDING FOR PROTECTION
In many of the world's at-risk cities, new structures are designed to survive earthquakes. However, for older structures, engineers need to innovate. These changes help to make the structures stronger, which will reduce damage and allow for quicker repairs.

WAVES OF DESTRUCTION
An earthquake's waves come in two forms. P-waves (yellow) arrive fastest and compress and punch the rock. S-waves (red) are slower but more destructive. They move from side to side to shake and destroy buildings. At the ground level, P- and S-waves combine to produce surface waves that crack windows and even destroy bridges.

翻譯：
地震：預測的挑戰
從洛杉磯、伊斯坦堡到東京和利馬，人類歷史上從未有過如此多的人口聚集在這些地震頻傳的城市 。這些城市位於地球巨大且移動的板塊邊緣，面臨著大地震帶來的嚴重破壞、經濟災難，以及隨之而來的海嘯、火災等破壞 。

與一個世紀前相比，我們對地震有了更多的了解 。科學家希望能夠預測地震，但這真的可能嗎？今天，一些關於地震的最簡單問題仍然難以回答：地震為何開始？又是什麼讓它停止？或許科學家最需要回答的問題是：地震是否存在清晰的模式，或者它們基本上是隨機的 (random) 且無法預測？

日本的預測研究
日本政府的科學家認為他們已經找到了答案 。東京大學地震研究所的科學家山岡耕春表示：「我們相信地震預測是可能的。」事實上，日本已經預測了下一次大地震的地點：東京西南方的東海地區 。這裡的兩個板塊邊界每 100 到 150 年就會產生一次大地震，但自 1854 年以來這裡就沒有發生過大地震 。理論認為，壓強正在這個區域 (zone) 累積，可能導致一場巨大的 (massive) 地震 。不幸的是，這更像是一種「預報」而非「預測」 。說一個高風險地區很可能發生地震是一回

事，但要精確預測地震發生的時間和地點則是另一回事 。

為了尋求時間與地點的精確 (precise) 預測，產生了另一個理論：「預滑 (preslip)」 。地震研究所的加藤尚之表示，他的實驗室 (laboratory) 實驗顯示，在地球地殼的斷層最終斷裂並引發地震之前，它會先發生微小的滑動 。如果我們能偵測 (detect) 到地殼深處發生的這些早期滑動，我們或許就能預測下一次大地震 。

沙漠中的線索

在加州帕克菲爾德（Parkfield）工作的科學家們也在嘗試研究預測地震的可能性 。他們選擇這個小鎮不僅是因為聖安德烈亞斯斷層穿過這裡，還因為這裡的地震相當規律——大約每 22 年一次 。1980 年代末期，科學家決定研究斷層，看看地震前是否有任何徵兆 。為此，他們鑽入斷層深處並設置設備來記錄活動，然後等待地震發生 。

年復一年，什麼也沒發生 。當地震終於在 2004 年 9 月 28 日發生時，比原定時程表 (schedule) 晚了數年，但最令人失望的是缺乏預警信號 。科學家審查了數據 (data)，卻找不到地震前有任何異常的證據 。這讓許多人相信地震或許真的是隨機事件 。然而，帕克菲爾德的科學家並未放棄，而是挖得更深 。到 2005 年夏末，他們抵達了斷層 3 公里的最終深度，在那裡繼續收集數據 (data)，希望能找到線索 。

終於，他們發現了某些東西 。在 2008 年 7 月的《自然》期刊發表的一篇文章中，研究人員聲稱在地震發生前不久偵測 (detect) 到了斷層的微小變化 。他們注意到了什麼？就在地震發生前，斷層的裂縫稍微變寬了 。科學家在芮氏規模 3.0 地震發生前 10 小時記錄到了第一波變化，並在規模 1.0 地震前兩小時發現了相同的跡象——這證明了「預滑」理論可能是正確的 。換句話說，預測地震事實上是有可能的 。

雖然還有很長的路要走，但從世界各地的研究來看，地震似乎並非完全隨機 。如果是這樣，未來我們或許能夠追蹤 (track) 地球的運動，並設計預警系統，讓我們能預測地震何時發生，進而防止生命損失 。

U4-A	
ancestor 祖先	a member of your family who lived a long time ago
Dawn 黎明	the period in the day when light from the sun begins to appear in the sky
destruction 破壞	摧毀 the act or process of destroying something
disaster 災難	an event which results in great harm, damage or death
displace 迫使遷離	強迫 to force something or someone out of its original position
expand 擴展	增加 to increase in size, number or importance
inevitable 不可避免的	防止 unable to be avoided or prevented
Monitor 監測	to watch a situation carefully for a period of time in order to discover something 情況 about it
summit 頂峰	the top of a mountain
witness 目擊	to see something happen, especially an accident or crime
U4-B	
data 數據	information that is collected
detect 察覺	to notice something that is partly hidden or not clear
foundation 地基	the solid layer of cement, bricks, stones etc. that is put under a building to support it
laboratory 實驗室	a room or building with scientific equipment for doing scientific tests
massive 巨大的	large in amount or degree
Precise 精確的	準確 exact and accurate
Random 隨機的	happening, done or chosen by chance
schedule 日程表	a list of planned activities showing the times or dates when they are intended to happen
Track 追蹤	記錄 進展 發展 to record the progress or development of something over a period of time

Zone 區域	an area used for different purposes than the area around it